
NetFix Unified Platform

Technical User Guide

Drive Test Intelligence • Root Cause Analysis • 5G Planning Intelligence

Purpose

This guide explains how optimization and planning engineers use the NetFix web platform: sign in, load data, run analysis, inspect evidence, use AI-assisted workflows, review geospatial results, and export engineering outputs.

NetFix Core Engineering Team

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Contents

1 Platform Overview and Role-Based Routing 2

1.1 Role-based routing 2

2 Drive Test Intelligence and Troubleshooting Portal 3

2.1 Navigation and workspace layout 3

2.2 Module 1: Telemetry Overview and ML Pipeline Dashboard 3

2.2.1 Top telemetry metrics 4

2.2.2 Severity distribution 4

2.2.3 Backend figure viewer 4

2.3 Module 2: Dataset Ingestion and Four-Stage Execution 4

2.3.1 Execution stages 5

2.4 Module 3: Deterministic RCA Engine and Telemetry Evidence 5

2.4.1 Incident selection and execution 6

2.4.2 Example rule match: MIMO Rank Fallback 6

2.4.3 AI engineering interpretation 6

2.4.4 23-metric telemetry evidence 6

2.4.5 Recommended actions and standards grounding 6

2.5 Module 4: Interactive GIS Trajectory Map 7

2.5.1 Controls and inspection workflow 7

2.6 Module 5: Penguin AI Copilot and Report Generator 7

2.6.1 Assistant controls and execution modes 8

2.6.2 Supported workflows 8

2.6.3 Context and telemetry sidebar 9

2.7 Module 6: Session History and Audit Trail 9

3 5G Network Planning Portal 11

3.1 Step 1: Sign in and open the Planning workspace 11

3.2 Step 2: Load a network-planning workbook (Upload & Plan) 12

3.3 Step 3: Review the network dashboard (Planning) 13

3.4 Step 4: Investigate conflicts (Clashes) 15

3.5 Step 5: Review assignments as a data table (Assignments) 16

3.6 Step 6: Add a new site or replan an existing one (Add Site) 17

3.6.1 Re-plan an existing site 17

3.6.2 Add a brand-new site 17

3.7 Step 7: Ask the OCTO copilot 18

3.8 Step 8: Compare planning revisions (Voronoi Compare) 20

3.9 Step 9: Export the deployment-ready workbook 21

3.10 Quick reference 21

4 Operational Notes 21

1 Platform Overview and Role-Based Routing

The **NetFix Unified Telecommunication Platform** brings two independent engineering workspaces into one web application:

Two engineering workspaces

Drive Test Intelligence / Anomaly Portal combines statistical anomaly detection, Histogram Gradient Boosting ML, PyTorch LSTM temporal validation, a 34-reason deterministic Root Cause Analysis (RCA) engine, interactive Leaflet GIS mapping, and the local **Penguin AI** assistant.

5G Network Planning Portal performs spatial geometry analysis, neighbor-graph validation, and constraint-based planning of 5G NR PCI and RSI while enforcing Mod3 and Mod4 conflict rules.

The datasets used by these workspaces are handled independently. The common platform layer provides authentication, role routing, user experience, persistence, reporting, and AI-assisted interaction.

1.1 Role-based routing

- **Optimization Engineer** → Drive Test Intelligence dashboard at `/dashboard`.
- **Planning Engineer** → 5G Network Planning Portal Upload & Plan workspace at `/np-upload`.

Quick start

Optimization workflow: Sign in → Upload dataset → Wait for the four pipeline stages → Review dashboard → Open an incident → Run RCA → Inspect map / Penguin AI / report.

Planning workflow: Sign in → Upload workbook → Inspect clashes → Bulk replan or add a site → Validate assignments → Export the final workbook.

2 Drive Test Intelligence and Troubleshooting Portal

Target user: Optimization Engineer / RF Performance Specialist

This section documents the operational workflow of the NetFix anomaly-detection and RCA portal, including navigation, telemetry monitoring, dataset ingestion, deterministic RCA, GIS inspection, AI-assisted analysis, reporting, and session history.

2.1 Navigation and workspace layout

The portal uses a persistent dark navigation sidebar and a light analysis workspace. The main entries are:

- **Dashboard** — telemetry overview, KPI cards, severity distribution, and backend validation figures.
- **Upload Dataset** — raw .pkl ingestion, four-stage progress tracking, and recent-session audit.
- **Results** — deterministic RCA, telemetry evidence, standards references, and recommended actions.
- **Interactive Map** — Leaflet GIS trajectory, cell/tower context, and anomaly markers.
- **Reports** — diagnostic summaries and exported analysis outputs.
- **Penguin AI** — local LLM engineering copilot for grounded analysis and report generation.
- **History** — historical sessions, completion states, anomaly counts, and cause-matching status.
- **Settings** — threshold defaults, model preferences, and API configuration.

The bottom-left user badge shows the authenticated engineer, current role, and logout controls.

2.2 Module 1: Telemetry Overview and ML Pipeline Dashboard

The Dashboard is the primary landing page after authentication. It gives an immediate view of the active dataset and provides direct access to generated backend figures (Figure 1).

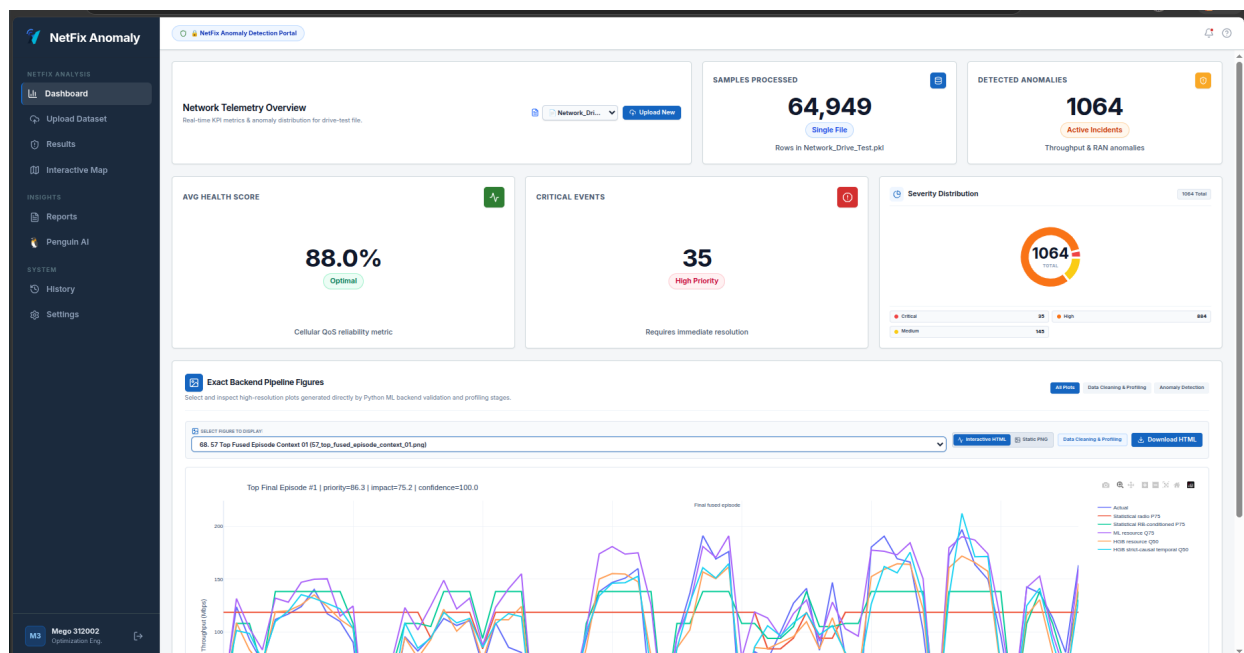


Figure 1: Telemetry dashboard with dataset metrics, incident severity distribution, and backend analysis figures.

2.2.1 Top telemetry metrics

Metric	Displayed value	Meaning
Samples Processed	64,949	Total telemetry rows loaded from <code>Network_Drive_Test.pkl</code> .
Detected Anomalies	1,064	Active incident episodes retained for RCA analysis and platform review.
Average Health Score	88.0%	Cellular QoS reliability indicator shown with an Optimal status.
Critical Events	35	Highest-priority operational cases requiring immediate engineering attention.

Table 1: Dashboard telemetry summary.

2.2.2 Severity distribution

The incident donut chart separates the 1,064 displayed incidents into:

- **Critical: 35** — severe throughput degradation or radio-link failure conditions.
- **High: 884** — significant degradation relative to expected performance.
- **Medium: 145** — moderate KPI variance or localized degradation.

2.2.3 Backend figure viewer

The figure panel exposes the exact plots produced by the Python analysis pipeline. Engineers can select an artifact, switch between interactive HTML and static PNG when available, and download the corresponding HTML output. Typical throughput-context figures compare:

- actual throughput;
- radio-conditioned statistical P75;
- RB-conditioned statistical P75;
- ML resource Q75;
- HGB resource-conditioned central prediction; and
- HGB strict-causal temporal central prediction.

2.3 Module 2: Dataset Ingestion and Four-Stage Execution

The Upload Dataset page controls ingestion and exposes pipeline status in real time (Figure 2).

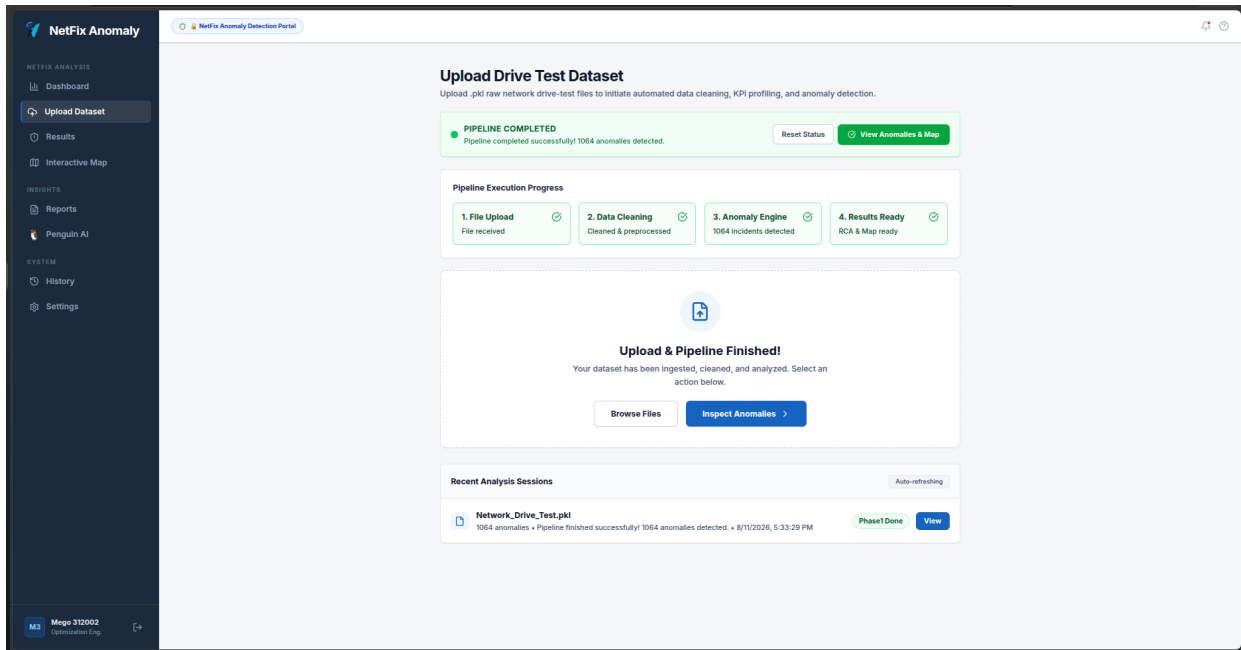


Figure 2: Dataset upload workflow with four-stage execution status and recent analysis sessions.

2.3.1 Execution stages

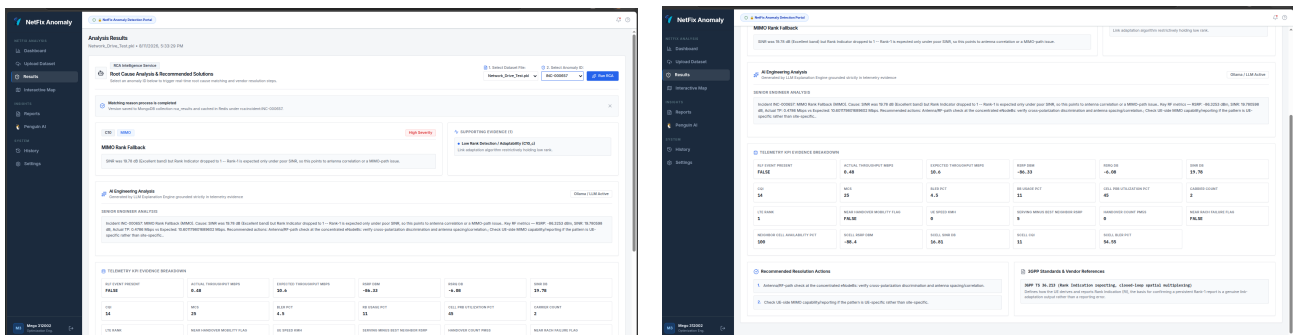
1. **File Upload** — validates the payload and receives the raw .pkl drive-test file.
2. **Data Cleaning** — executes the 11-stage non-leaking preprocessing pipeline (**ran-cleaning-service**).
3. **Anomaly Engine** — runs the 19-stage statistical, HGB ML, and PyTorch LSTM anomaly-fusion workflow.
4. **Results Ready** — builds MongoDB session indexes and prepares RCA cause matching and GIS artifacts.

On success, the page confirms pipeline completion and the incident count. **View Anomalies & Map** opens the investigation workspace; **Reset Status** prepares the interface for a new upload.

The Recent Analysis Sessions table records dataset name, anomaly count, timestamp, pipeline state, and a direct session-view action.

2.4 Module 3: Deterministic RCA Engine and Telemetry Evidence

The Results workspace is used to select an incident, run deterministic RCA, inspect telemetry evidence, review AI-supported engineering interpretation, and follow recommended actions. Figures 3a and 3b show the two main views.



(a) Matched RCA rule, AI engineering analysis, and telemetry evidence grid.

(b) Recommended actions and 3GPP standards reference.

Figure 3: Deterministic RCA workflow for an individual incident.

2.4.1 Incident selection and execution

Select the dataset and Incident ID (for example, INC-000657), then click **Run RCA**. The completion banner confirms that the matched-reason process has finished, saves the result to MongoDB collection `rca_results`, and caches the incident in Redis under the corresponding incident key.

2.4.2 Example rule match: MIMO Rank Fallback

For INC-000657, the portal shows:

- **Category / severity:** C10, MIMO, High Severity.
- **Matched reason:** MIMO Rank Fallback.
- **Reasoning:** SINR is 19.78 dB while Rank Indicator falls to 1; Rank-1 under strong SINR suggests antenna correlation or a MIMO-path problem rather than weak RF conditions.
- **Supporting evidence:** low-rank adaptation remains active despite clean RF conditions.

2.4.3 AI engineering interpretation

The local Ollama analysis card combines the deterministic rule with the actual telemetry. In the illustrated incident it reports RSRP = -86.3253 dBm, SINR = 19.780598 dB, actual throughput = 0.4786 Mbps, and expected throughput = 10.601179 Mbps. The recommended engineering checks focus on antenna / RF-path behavior, cross-polarization discrimination, spacing / correlation, and UE-side MIMO capability when the pattern appears UE-specific.

2.4.4 23-metric telemetry evidence

Telemetry field	Recorded value	Evaluation
RLF EVENT PRESENT	FALSE	No Radio Link Failure event.
ACTUAL THROUGHPUT MBPS	0.48 Mbps	Severe degradation; expected throughput is 10.6 Mbps.
EXPECTED THROUGHPUT MBPS	10.6 Mbps	Calculated P75 baseline for the cell/context.
RSRP DBM	-86.33 dBm	Good serving-cell signal strength.
RSRQ DB	-6.08 dB	Good serving-cell signal quality.
SINR DB	19.78 dB	Excellent signal-to-interference ratio.
CQI	14	High Channel Quality Indicator.
MCS	25	High Modulation and Coding Scheme index.
BLER PCT	4.5%	Low Block Error Rate.
RB USAGE PCT	11%	Low Resource Block usage.
CELL PRB UTILIZATION PCT	45%	Moderate cell load.
CARRIER COUNT	2	Dual component carrier; CA active.
LTE RANK	1	Anomalous under 19.78 dB SINR.
NEAR HANDOVER MOBILITY FLAG	FALSE	Stationary / low-mobility state.
UE SPEED KMH	0 km/h	Stationary UE.
SERVING MINUS BEST NEIGHBOR RSRP	5 dB	Adequate cell-dominance margin.
HANDOVER COUNT PM5S	0	No recent handovers.
NEAR RACH FAILURE FLAG	FALSE	Random-access behavior normal.
NEIGHBOR CELL AVAILABILITY PCT	100%	All neighbor cells available.
SCELL RSRP DBM	-88.4 dBm	Secondary-cell RSRP clean.
SCELL SINR DB	16.81 dB	Secondary-cell SINR clean.
SCELL CQI	11	Secondary-cell CQI acceptable.
SCELL BLER PCT	54.55%	Secondary-cell BLER elevated.

Table 2: Telemetry evidence shown for the example MIMO Rank Fallback incident.

2.4.5 Recommended actions and standards grounding

1. Check the antenna / RF path at concentrated eNodeBs, including cross-polarization discrimination and antenna spacing / correlation.
2. Check UE-side MIMO capability and reporting if the behavior is UE-specific rather than site-specific.

- Review **3GPP TS 36.213** for Rank Indication reporting and closed-loop spatial multiplexing; this standard defines how the UE derives and reports RI.

2.5 Module 4: Interactive GIS Trajectory Map

The Interactive Map renders the drive-test route and incident locations on a Leaflet canvas (Figure 4).

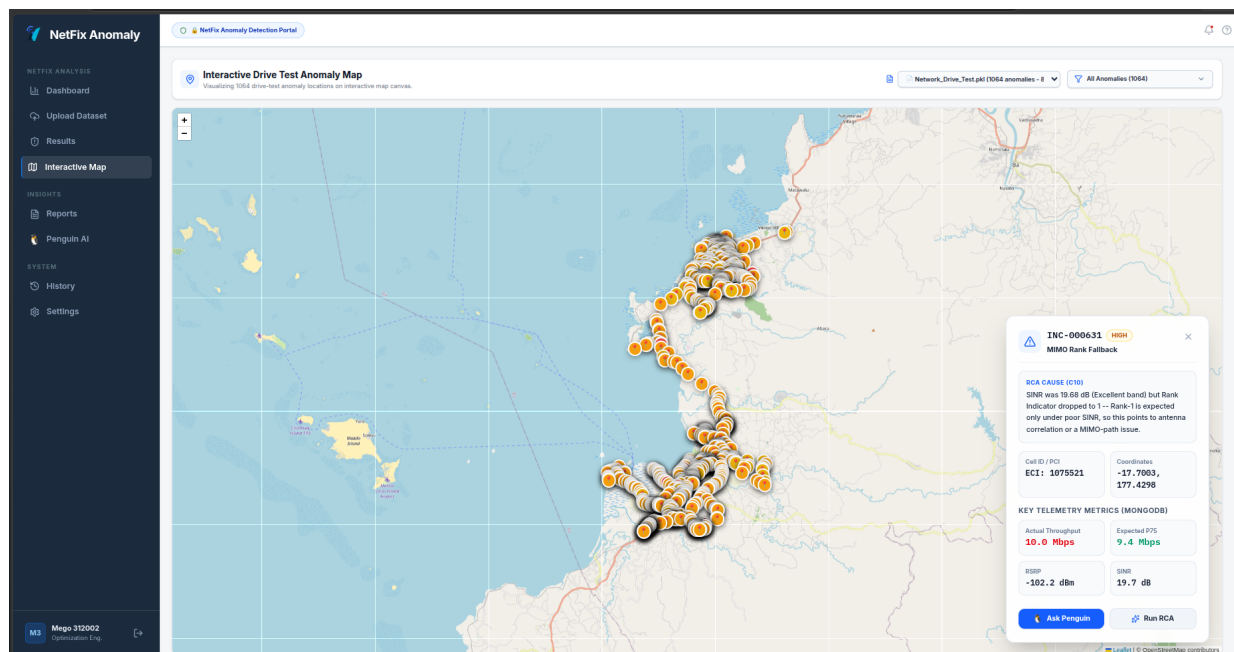


Figure 4: Drive-test trajectory map with anomaly markers and an incident detail popup.

2.5.1 Controls and inspection workflow

- **Dataset selector** — choose the active drive-test dataset.
- **Incident filter** — display all 1,064 incidents or a selected severity tier.
- **Map navigation** — pan and zoom over OpenStreetMap tiles.
- **Incident marker** — open an incident card with severity, reason, serving-cell metadata, coordinates, and selected KPI values.

For the illustrated incident **INC-000631**, the popup identifies a high-severity **MIMO Rank Fallback** case. The displayed context includes Cell / PCI **ECI: 1075521**, coordinates **-17.7003, 177.4298**, actual throughput **10.0 Mbps**, expected P75 **9.4 Mbps**, RSRP **-102.2 dBm**, and SINR **19.7 dB**. **Ask Penguin** opens the assistant with the incident context preloaded; **Run RCA** opens the detailed RCA workspace.

2.6 Module 5: Penguin AI Copilot and Report Generator

Penguin AI provides a local conversational interface for grounded telecom analysis, incident explanation, KPI relationship reasoning, and structured report generation. Figures 5–7 illustrate the main interaction modes.

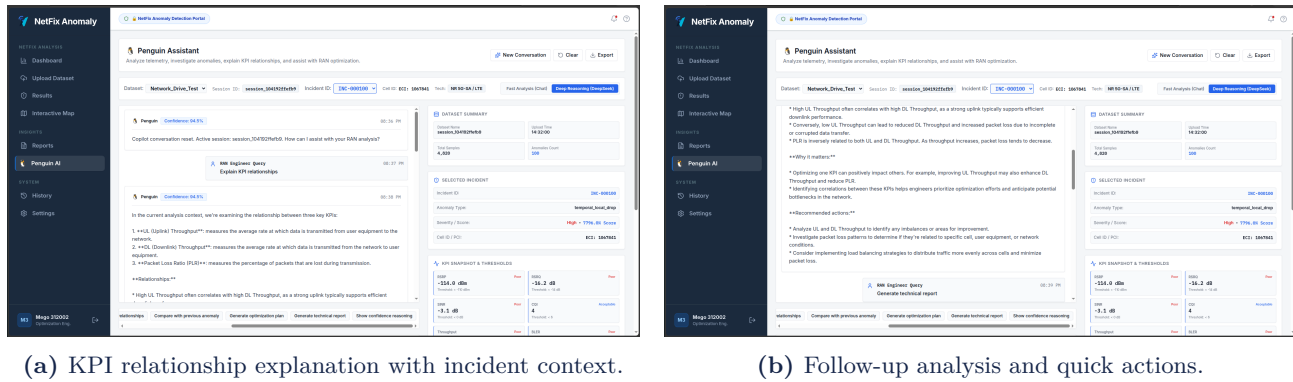


Figure 5: Penguin AI conversational engineering workflow.



Figure 6: Automated technical-report generation sequence.

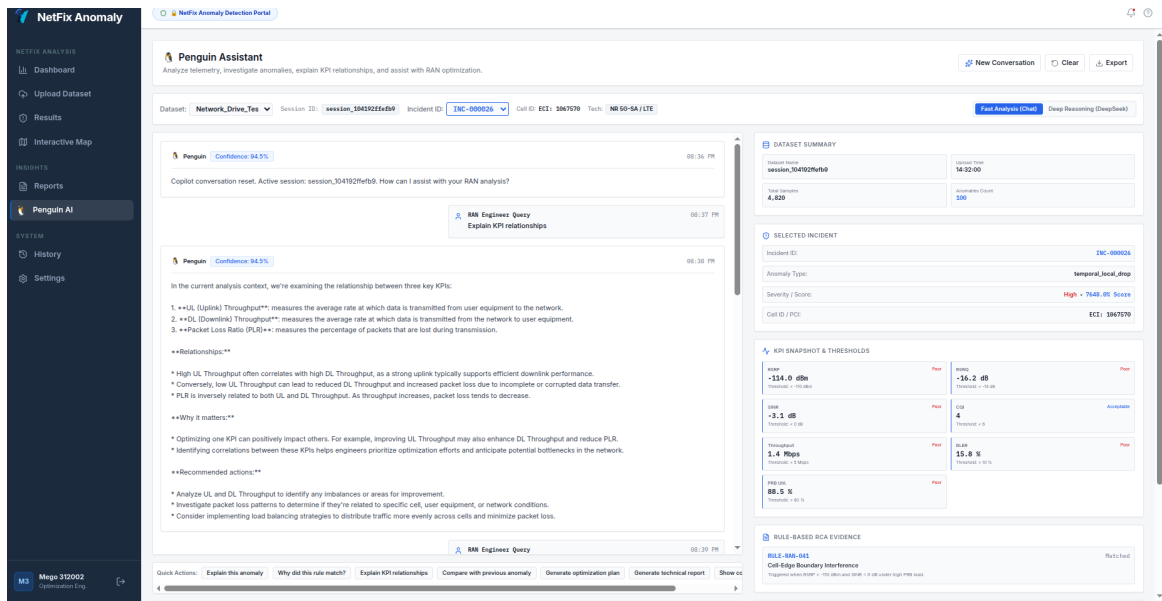


Figure 7: Penguin AI analyzing an incident matched to RULE-RAN-041 Cell-Edge Boundary Interference.

2.6.1 Assistant controls and execution modes

- **Conversation controls:** New Conversation, Clear, and Export.
- **Context bar:** active dataset, session, incident, cell, and radio technology. The illustrated context includes `Network_Drive_Test`, `session_104192ffefb9`, `INC-000100 / INC-00026`, `ECI: 1067841`, and `NR 5G-SA / LTE`.
- **Fast Analysis (Chat):** Llama 3.1 8B for direct engineering dialogue.
- **Deep Reasoning (DeepSeek):** DeepSeek R1 8B for extended reasoning workflows.

2.6.2 Supported workflows

- **Explain KPI relationships** — in the illustrated example, Penguin reports 94.5% confidence and explains the interaction among UL throughput, DL throughput, and Packet Loss Ratio

(PLR), including the effect of delayed TCP ACKs and TCP-window reduction on DL performance.

- **Generate technical report** — creates session metadata, executive summary, evidence table, RCA, recommended actions, and confidence assessment. The sample report identifies `session_104192ffefb9`, date `2023-02-20`, and author `Penguin, AI Engineering Copilot`; it highlights Low UL Throughput (`-2.1 dBm`), High PLR (`4.7%`), and Moderate DL Throughput (`-0.8 dBm`), then recommends UL optimization, PLR/load-balancing actions, and post-remediation DL verification.
- **Explain this anomaly / Why did this rule match?** — grounds the answer in the active incident and deterministic RCA evidence.
- **Compare with previous anomaly / Generate optimization plan / Show confidence reasoning** — provides one-click investigation prompts.

2.6.3 Context and telemetry sidebar

The right-side context panel keeps the active data and engineering thresholds visible while the assistant responds. The illustrated dataset summary shows **4,820 samples** and **100 anomalies**; the selected-incident card displays `temporal_local_drop`, Cell ECI: `1067841`, and the UI severity / score string `High 7796.0% Score`.

KPI	Displayed value	Displayed threshold / state
RSRP	-114.0 dBm	Poor; threshold < -110 dBm
RSRQ	-16.2 dB	Poor; threshold < -14 dB
SINR	-3.1 dB	Poor; threshold < 0 dB
CQI	4	Acceptable; threshold < 6
Throughput	1.4 Mbps	Poor; threshold < 5 Mbps
BLER	15.8%	Poor; threshold > 10%
PRB utilization	88.5%	Poor; threshold > 80%

Table 3: Example KPI snapshot and thresholds displayed beside Penguin AI.

The same sidebar displays the matched deterministic RCA evidence, for example `RULE-RAN-041 Cell-Edge Boundary Interference`, so the LLM response remains connected to visible engineering evidence.

2.7 Module 6: Session History and Audit Trail

The History page maintains an audit trail of platform analysis sessions (Figure 8).

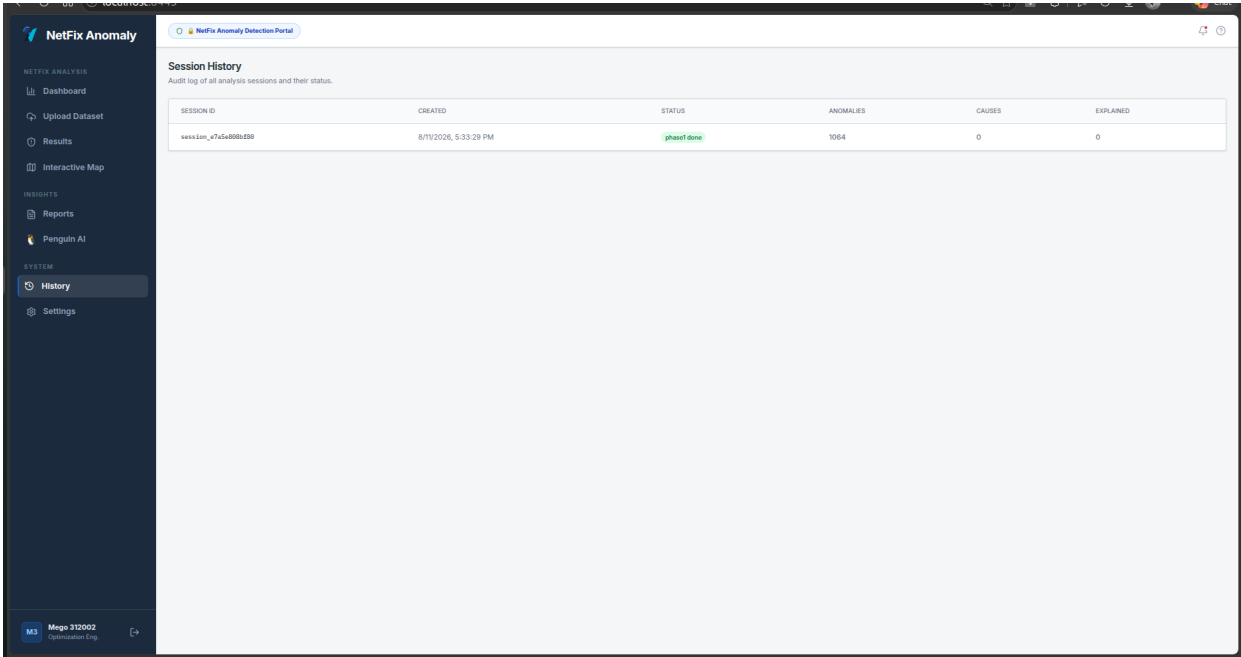


Figure 8: Session History showing timestamps, processing status, anomaly totals, and RCA progress.

The table records **SESSION ID**, **CREATED**, **STATUS**, **ANOMALIES**, **CAUSES**, and **EXPLAINED**. For example, a completed Phase-1 session may show 1,064 anomalies before subsequent cause-matching and explanation fields are populated.

3 5G Network Planning Portal

Target user: 5G RF Planning Engineer / Network Deployment Specialist

The 5G Network Planning Portal supports workbook ingestion, clash auditing, automated bulk replanning, interactive site addition, assignment review, before/after Voronoi comparison, AI-assisted network Q&A through the OCTO copilot, and export of deployment-ready planning parameters.

Every antenna on every tower in a 5G network must be assigned a small set of unique identifiers (PCI, RSI, PRACH root sequences) so that user equipment does not confuse nearby cells. The 5G Network Planning Portal automates that assignment, renders the full network on an interactive map, flags every clash it detects, and lets the engineer replan, add sites, and export the final workbook.

3.1 Step 1: Sign in and open the Planning workspace

1. Open the portal in the browser and enter the engineer credentials.
2. Select **5G Network Planning** from the workspace selector at the top of the sign-in form (the alternative *Drive Test & Anomaly* workspace routes to the Optimization dashboard covered in Section 2).
3. Click **Sign In**. The application routes the Planning Engineer to the **Upload & Plan** screen.

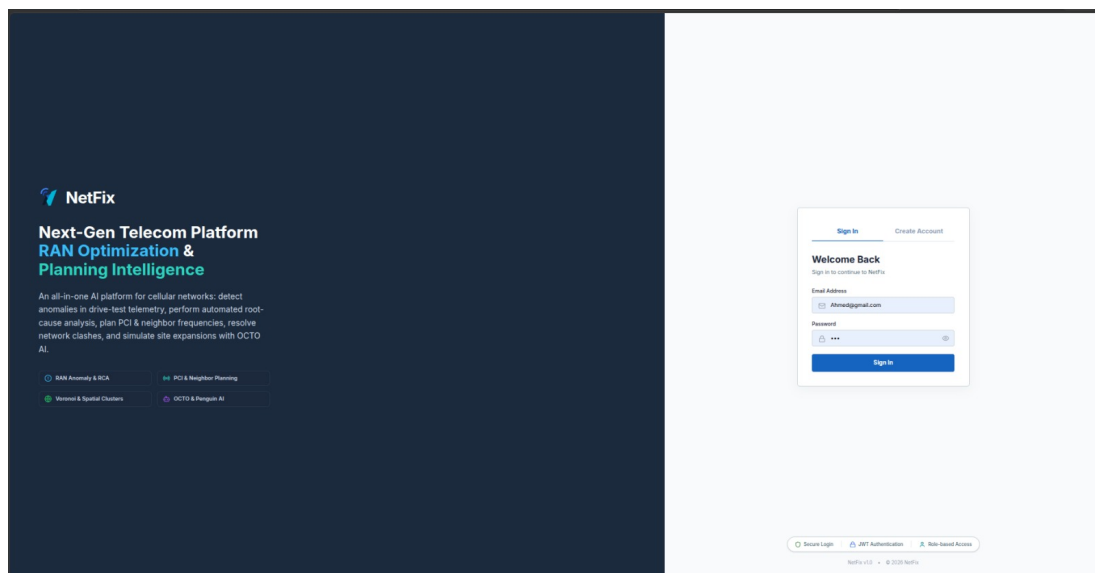


Figure 9: Sign-in screen with the 5G Network Planning workspace selected.

Once authenticated, a persistent left sidebar exposes every section of the workspace: Upload & Plan, Planning, Clashes, Assignments, Add Site, OCTO, and Voronoi Compare.

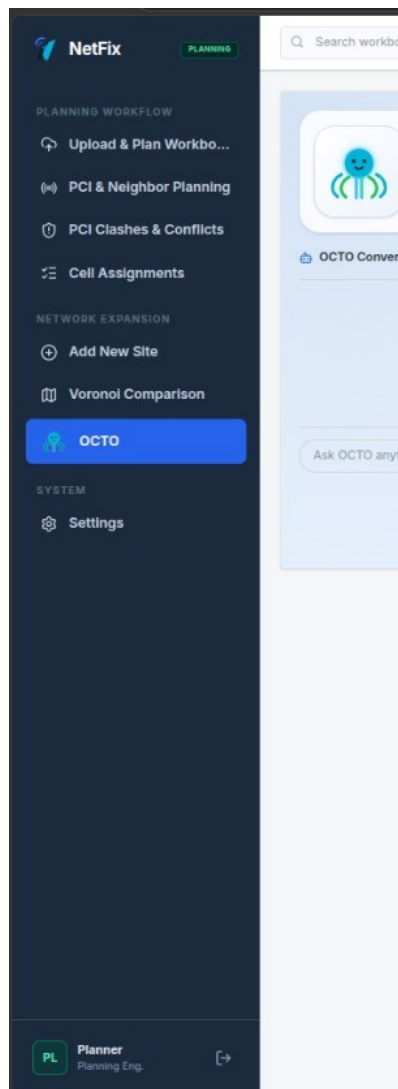


Figure 10: Sidebar of the 5G Network Planning Portal showing every available section.

3.2 Step 2: Load a network-planning workbook (Upload & Plan)

The Upload & Plan page is the mandatory entry point — no other tab is enabled until a workbook has been loaded. Two ingestion modes are supported through separate tabs at the top of the page:

- **Raw** — for workbooks that list sites, sectors, and antennas but where PCI / RSI have not yet been assigned. After uploading, click **Run** to let the constraint solver assign all identifiers automatically. A staged progress indicator reports each phase of the pipeline. On completion, the page shows a pass/fail summary and enables the download of the completed workbook.
- **Final** — for workbooks that already contain a full set of assignments. Click **Use workbook** to load it directly for review, clash auditing, and exploration; no planning step is required.

The same page also includes a built-in **column reference guide** explaining every required column in plain language, with an example value for each, and a button to download a blank template.

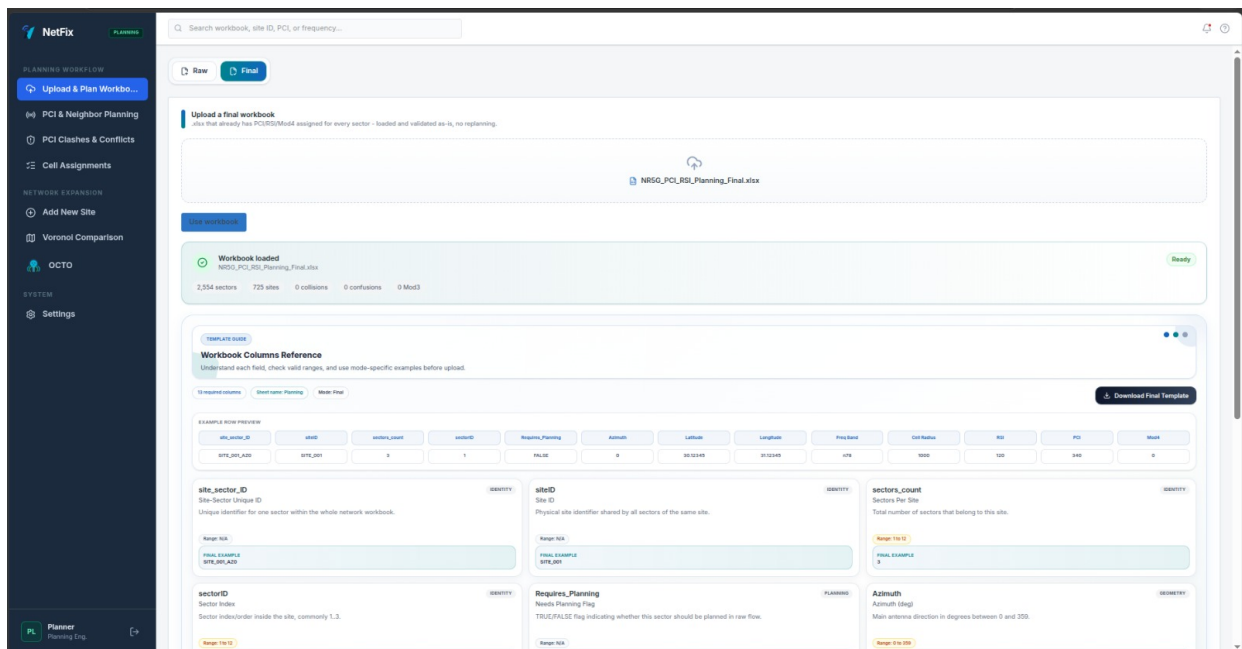


Figure 11: Upload & Plan screen showing both Raw and Final ingestion tabs along with the built-in column reference guide and template download.

Prerequisite

Every other tab in the sidebar remains disabled until a workbook has been successfully loaded on this page. Switching tabs beforehand prompts the user to return to Upload & Plan first.

3.3 Step 3: Review the network dashboard (Planning)

The **Planning** tab is the top-level health view for the loaded network. At the top of the page, a global **PASS/FAIL** status card is accompanied by a count of serious issues. Immediately below, a detailed breakdown splits every detected clash into two categories:

- **Hard rules** (red) — genuine conflicts that must be resolved (for example PCI collisions, Mod4 clashes, RSI collisions).
- **Soft rules** (amber) — minor overlaps that are expected in dense deployments and are shown for information only (for example Mod3 or Mod30 warnings).

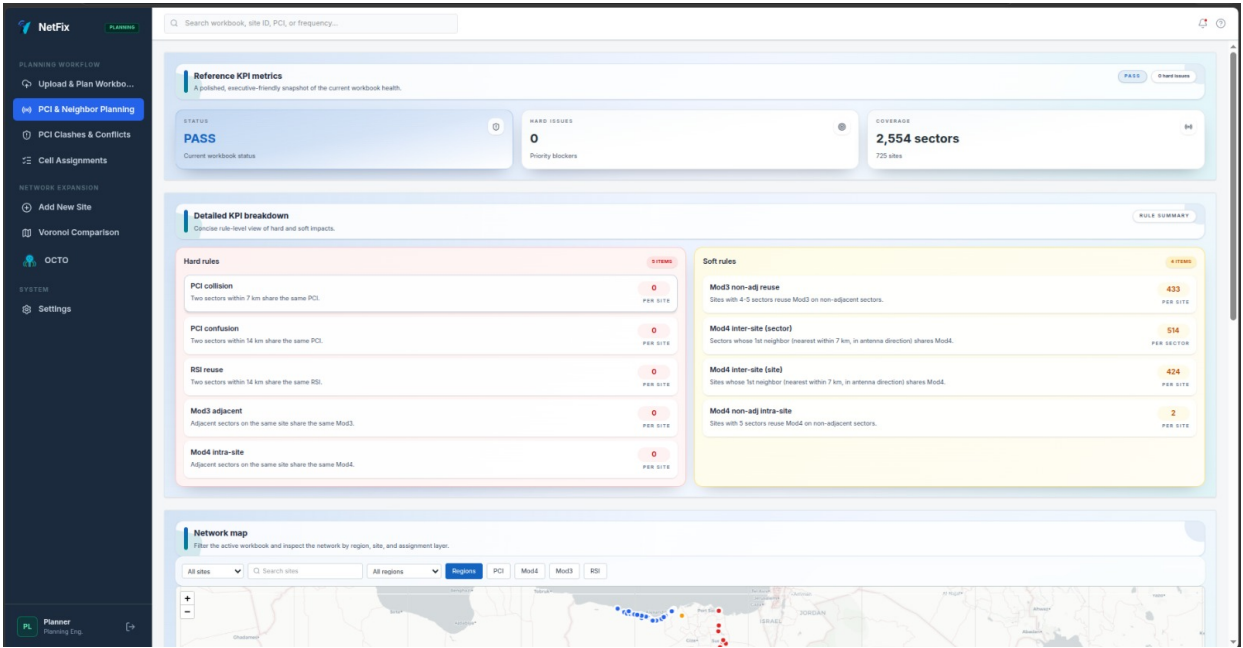


Figure 12: Planning dashboard showing overall PASS/FAIL status and the detailed hard-rule / soft-rule breakdown.

Below the KPI cards, an interactive map renders the full network. Engineers can:

- Search for a specific site or filter by region using the dropdowns above the map.
- Switch the map colouring (by region, PCI, RSI, or Mod-group) using the row of buttons above the map.
- Pan by dragging and zoom by scrolling.

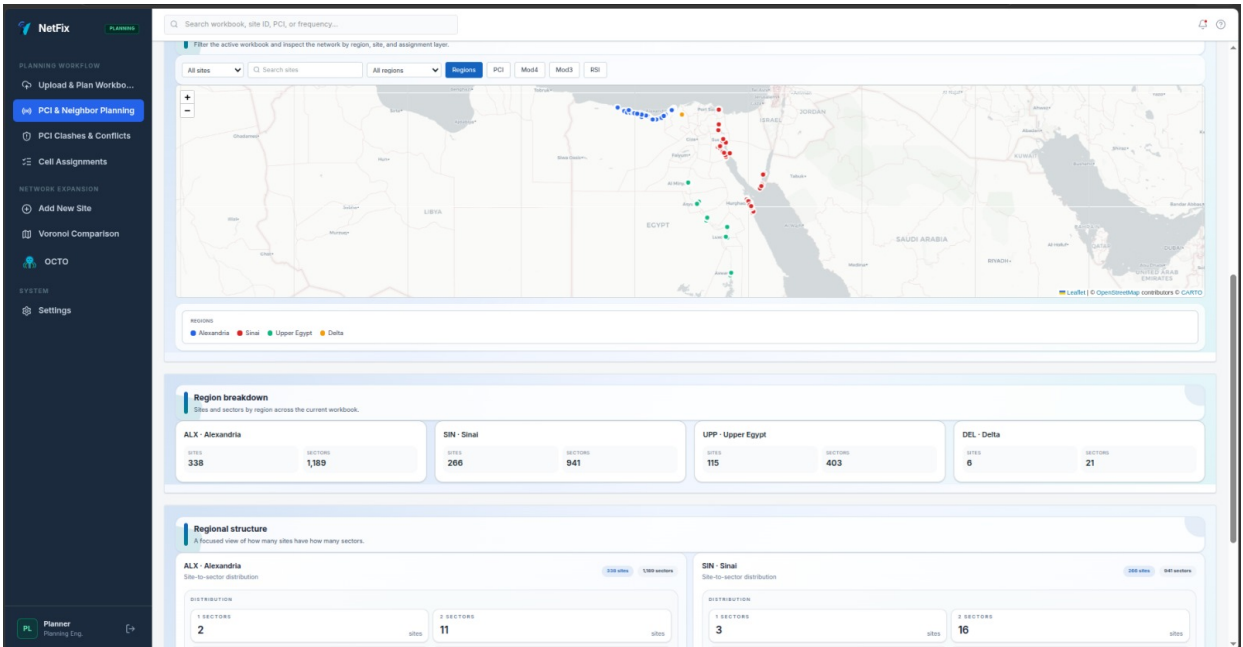


Figure 13: Interactive network map with region / site filters and colour-mode selectors.

The bottom of the page contains a region-by-region breakdown showing the number of sites and sectors present in each region of the network.

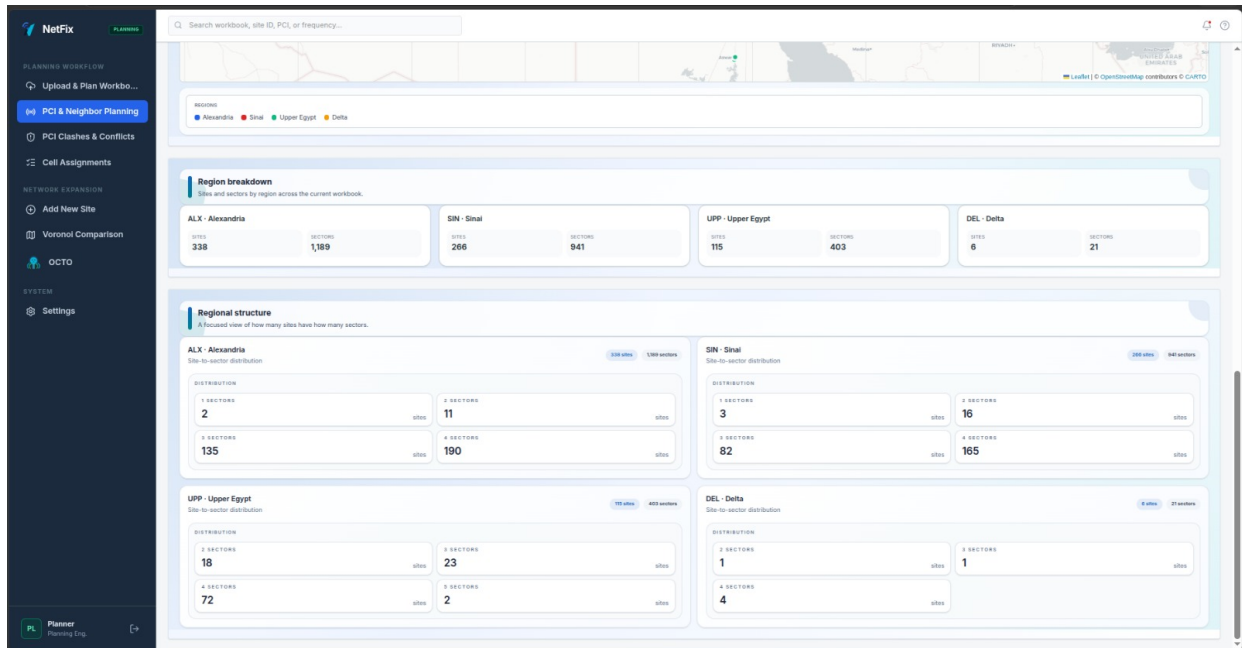


Figure 14: Per-region breakdown cards at the bottom of the Planning tab.

3.4 Step 4: Investigate conflicts (Clashes)

The **Clashes** tab is where every issue flagged on the Planning dashboard is investigated. The map on this tab looks similar to the Planning map, but every sector is coloured by *clash type* rather than by region. Coloured filter chips above the map toggle individual clash categories on and off, and a **Colour sectors by** selector on the right switches the map shading mode.

Clicking any sector on the map opens a right-hand panel with three modes:

- **Measure** — displays every neighbouring site inside an adjustable radius; useful for understanding why two sectors have been flagged as too close.
- **Explainer** — accepts any sector / antenna identifier and returns a short readable summary of whether the sector has an issue and exactly which neighbour is causing it.
- **Clash Inspector** — opens below the map when a sector wedge is clicked, listing every rule the sector is breaking with a plain-language justification.

A “**Show full clash reference**” link below the map expands a glossary of every clash type the engine can detect, with a short plain-English explanation of each.

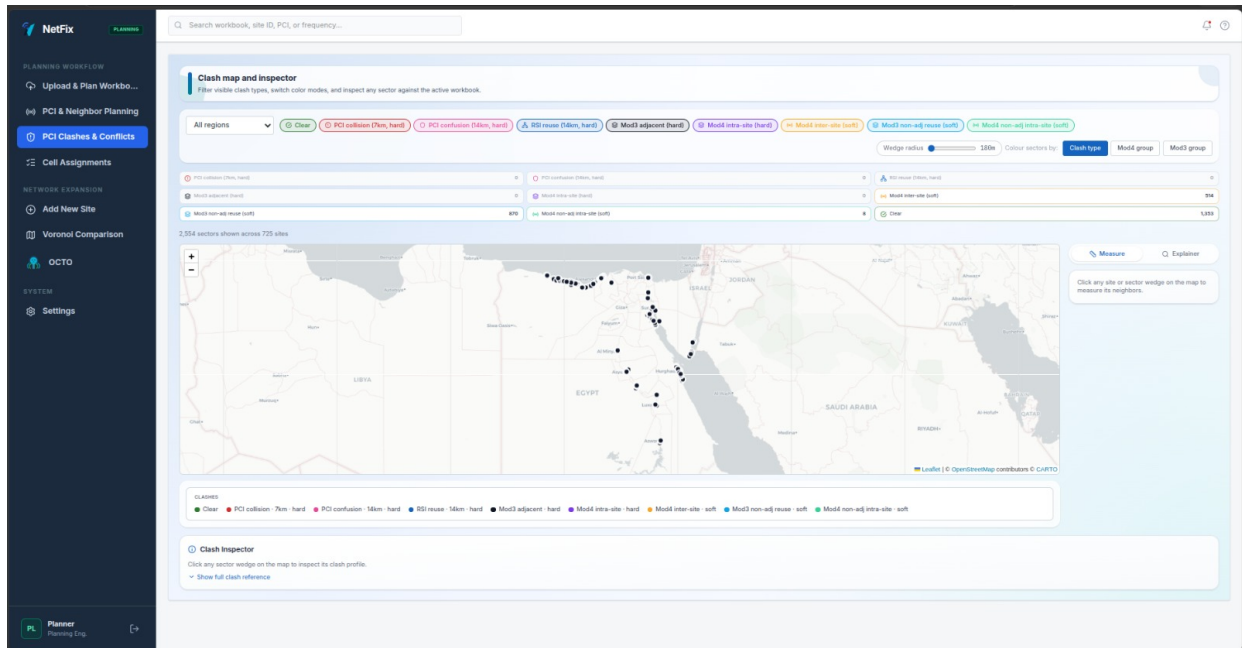


Figure 15: Clashes workspace: colour-coded conflict map, filter chips, sector inspection panel, and expandable clash reference glossary.

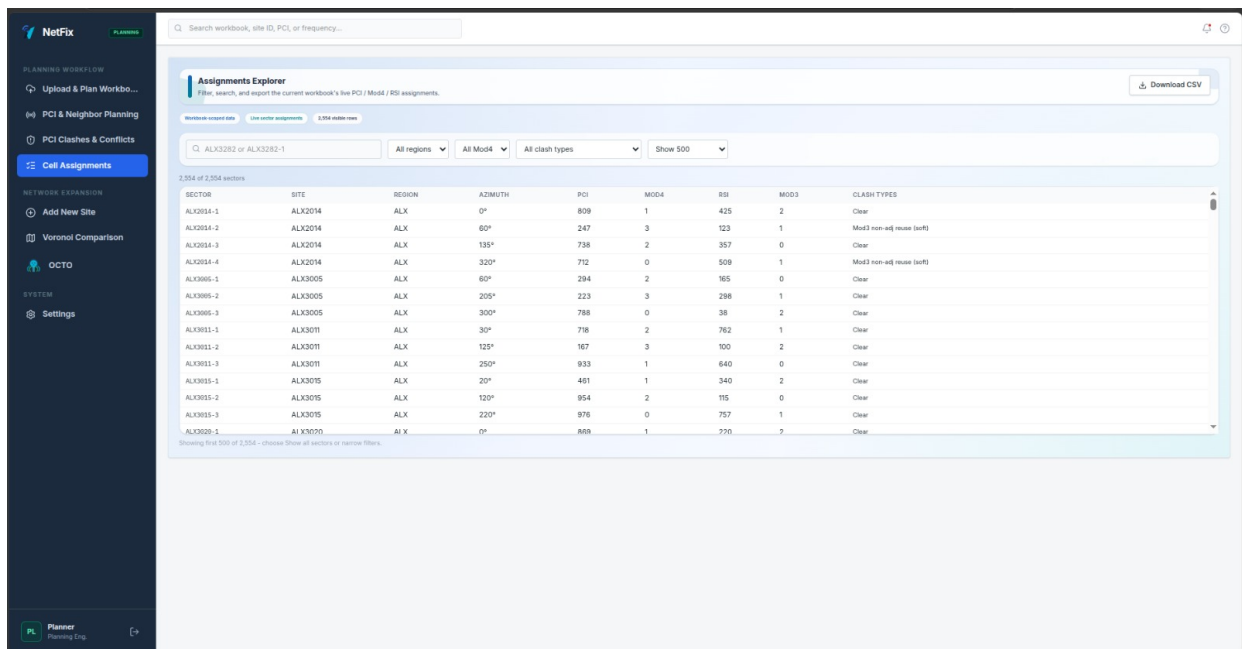
The four principal clash categories exposed on this page are:

- **PCI collision (hard):** same PCI assigned to adjacent sectors within the configured reuse distance.
- **Mod4 clash (hard):** same Modulo-4 index causing DMRS sequence overlap.
- **Mod3 / Mod30 warnings (soft):** PSS / SSS symbol-shift conflicts.
- **RSI collision (hard):** PRACH root-sequence contention.

3.5 Step 5: Review assignments as a data table (Assignments)

The **Assignments** tab exposes every sector in the loaded network as rows in a searchable, sortable table — useful when working directly with tabular data rather than the map.

- Use the search box to locate a site or sector by name.
- Use the dropdown filters to restrict the view by region, group index, or clash type.
- Click any column heading to sort by that column.
- Use the **Download CSV** button at the top right to export exactly the currently-visible view (filters included) as a spreadsheet.



Assignments Explorer
Filter, search, and export the current workbook's live PCI / Mod4 / RSI assignments.

Reverse sort order | Live sector assignments | 5/100 table rows

Search: ALX3202 or ALX3202-1 | All regions | All Mod4 | All clash types | Show 500

SECTOR	SITE	REGION	AZIMUTH	PCI	MOD4	RSI	MOD3	CLASH TYPES
ALX2014-1	ALX2014	ALX	0°	809	1	425	2	Clear
ALX2014-2	ALX2014	ALX	60°	247	3	123	1	Mod3 non-adj reuse (soft)
ALX2014-3	ALX2014	ALX	135°	738	2	357	0	Clear
ALX2014-4	ALX2014	ALX	320°	712	0	508	1	Mod3 non-adj reuse (soft)
ALX3005-1	ALX3005	ALX	60°	294	2	165	0	Clear
ALX3005-2	ALX3005	ALX	205°	223	3	298	1	Clear
ALX3005-3	ALX3005	ALX	300°	788	0	38	2	Clear
ALX3011-1	ALX3011	ALX	30°	718	2	762	1	Clear
ALX3011-2	ALX3011	ALX	125°	167	3	100	2	Clear
ALX3011-3	ALX3011	ALX	250°	933	1	640	0	Clear
ALX3015-1	ALX3015	ALX	20°	481	1	340	2	Clear
ALX3015-2	ALX3015	ALX	120°	954	2	115	0	Clear
ALX3015-3	ALX3015	ALX	220°	976	0	757	1	Clear
ALX3020-1	ALX3020	ALX	0°	808	1	770	2	Clear

Showing first 500 of 2,554 - choose Show all sectors or narrow filters.

Figure 16: Assignments table with search, filters, sortable columns, and CSV export.

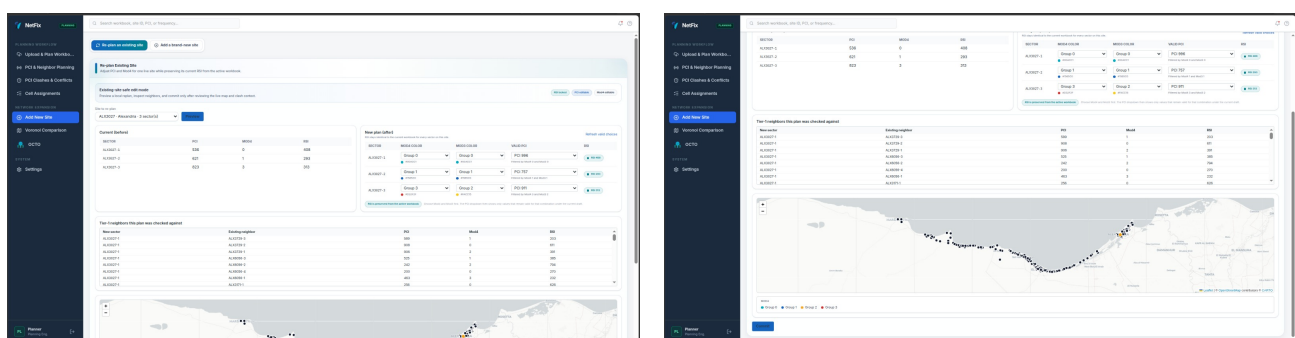
3.6 Step 6: Add a new site or replan an existing one (Add Site)

The **Add Site** tab supports two workflows for modifying the network.

3.6.1 Re-plan an existing site

Use this workflow when a site already exists in the loaded network but requires re-checking or re-assignment of its identifiers.

1. Select the **Re-plan an existing site** option at the top of the page.
2. Choose the target site from the dropdown and click **Preview**.
3. Review the side-by-side comparison of current versus proposed parameter assignments, together with a small map of the site and its immediate neighbours.
4. Click **Commit** to persist the replanned assignment.



(a) Site selection and proposed replanned assignments.

(b) Map review of the replanned site and its immediate neighbours.

Figure 17: Re-plan an existing site workflow.

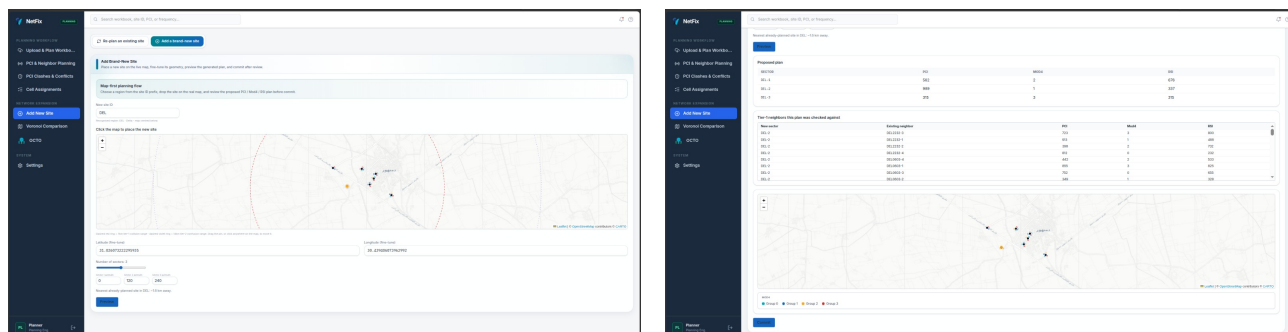
3.6.2 Add a brand-new site

Use this workflow when the target site does not yet exist in the network.

1. Select the **Add a brand-new site** option at the top of the page.
2. Enter a name for the new site — the map recentres on the appropriate region as soon as the name is recognised.
3. Click on the map to place the new site, or drag the pin to fine-tune its position. Coordinates

may also be entered manually.

4. Configure the number of sectors and their azimuths using the slider and numeric fields (typical trisector configuration: 0° , 120° , 240°).
5. Click **Preview** to generate valid PCI and RSI candidate assignments and to visualise how the new site fits its neighbours.
6. Click **Commit** to add the site to the network permanently.



(a) Placing the new site on the interactive map picker. (b) Preview of candidate assignments before committing.

Figure 18: Add a brand-new site workflow.

3.7 Step 7: Ask the OCTO copilot

The **OCTO** tab exposes an in-workspace **AI agent** that answers plain-English questions grounded on the currently loaded network. Unlike a plain chatbot, OCTO follows an agentic loop:

1. **User input** — the engineer types a natural-language question (for example, “How is the Sinai region doing?”).
2. **LLM reasoning** — the underlying large language model interprets the question, decides which piece of network information is needed, and selects the most suitable *tool* from its registered toolset.
3. **Tool invocation** — the selected tool is executed against the currently loaded network (for example, a site lookup, distance query, clash inspection, or regional summary), returning structured data.
4. **LLM answer generation** — the LLM takes the raw output returned by the tool and composes a plain-English, engineer-friendly answer for the user.

Questions typical of a planning engineer include:

- “Why does site ALX3282 have a problem?”
- “How is the Sinai region doing overall?”
- “What are the five worst sites in my network?”
- “How far apart are these two sites?”

Whenever OCTO invokes a tool to answer, it displays an expandable *tool call* chip in the conversation so the engineer can inspect exactly which tool was called, what arguments were passed, and what raw data was returned — keeping every AI-generated answer fully transparent and auditable. A right-hand **Toolset** panel lists every capability the agent can invoke, grouped by category (site lookup, clash inspection, distance queries, regional summaries, ranking, etc.).

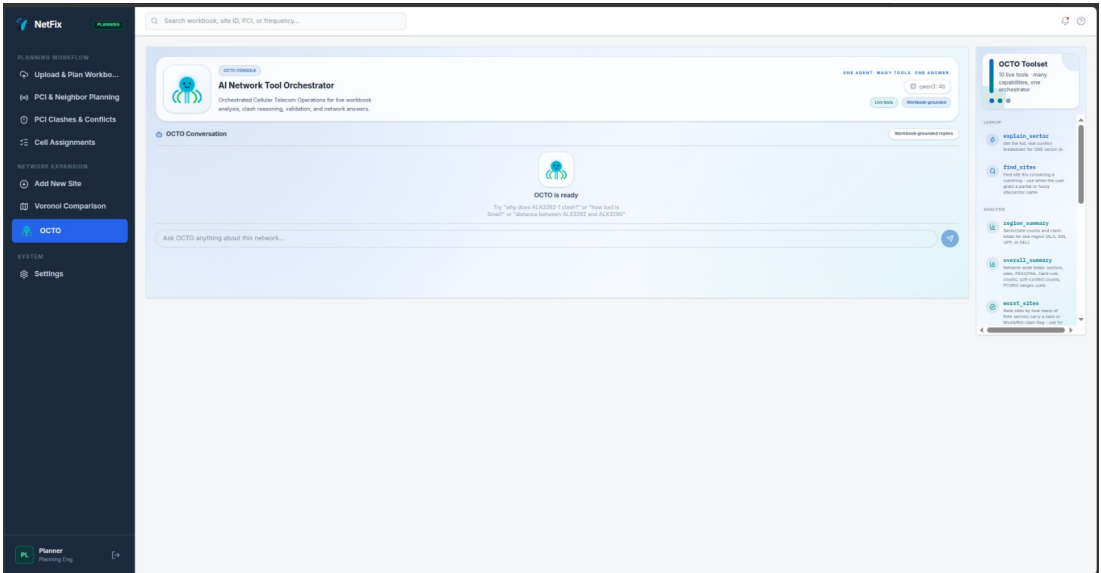


Figure 19: OCTO conversation view with an expandable tool-call chip showing which tool the LLM invoked to answer the question.

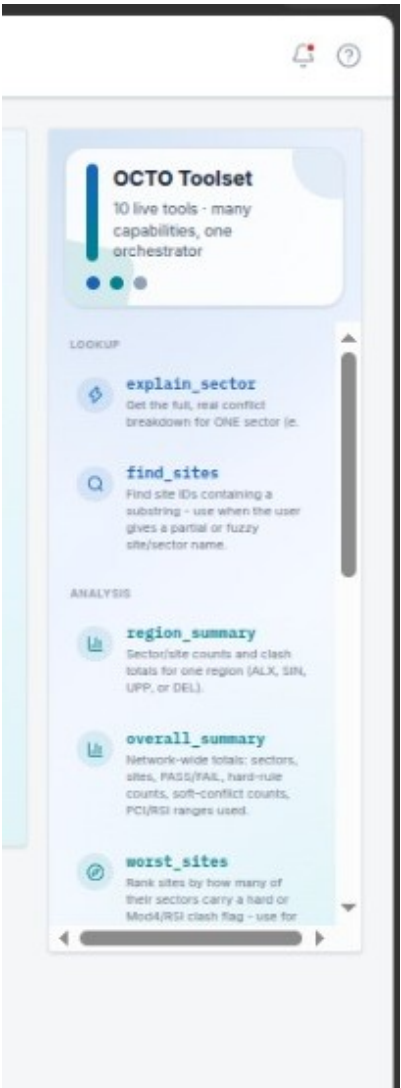


Figure 20: OCTO Toolset panel listing every capability the AI agent can invoke.

Scope of OCTO

OCTO is a *read-only* AI agent — it can only invoke tools that look up and analyze the currently loaded network. It does not perform planning, modify assignments, or commit changes; all planning and commit actions remain explicit engineer operations performed on the Upload & Plan or Add Site tabs.

3.8 Step 8: Compare planning revisions (Voronoi Compare)

The **Voronoi Compare** tab lets engineers compare an older revision of the network against the currently loaded version, focused on a specific site and its neighbourhood.

1. Upload the older **.xlsx** revision using the upload box at the top of the tab.
2. Select the site of interest and choose how many neighbouring sites to include in the comparison using the dropdown and slider.
3. Switch between **Diagram** view (simplified, colour-coded Voronoi cells) and **Real map** view (accurate map with real sector positions) using the toggle.
4. A status badge at the top of the comparison reports the outcome as **Improved**, **Fully Solved**, **Unchanged**, or **Regressed**.

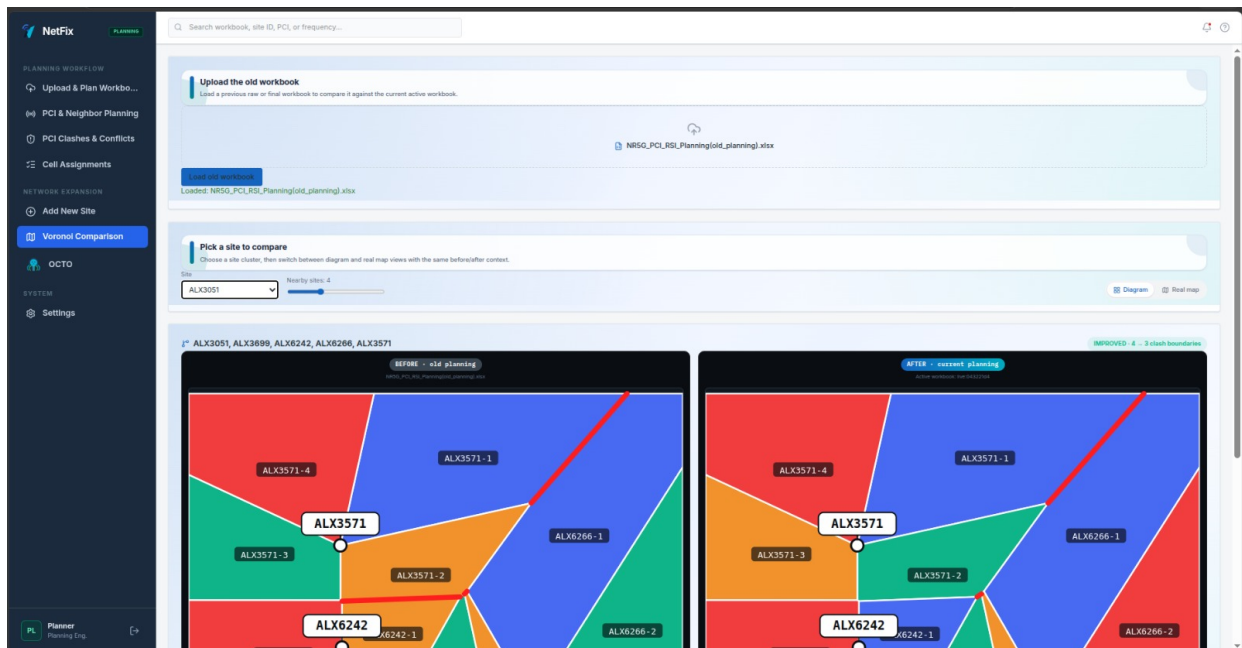
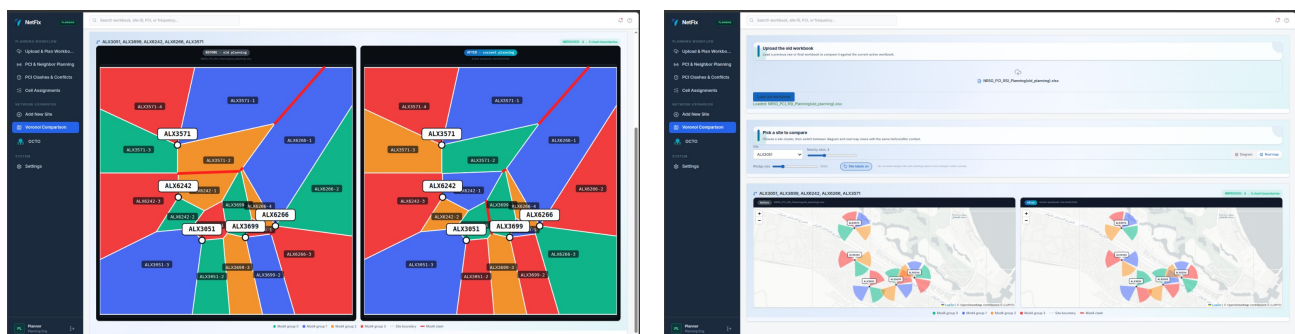


Figure 21: Voronoi Compare: selecting the focus site and comparison scope.



(a) Diagram view: before / after comparison.

(b) Real map view: before / after comparison.

Figure 22: Voronoi Compare workflow visualising the impact of a replanning revision.

3.9 Step 9: Export the deployment-ready workbook

At any point once the network has been loaded and validated, use the **Download** action on the Upload & Plan tab (also available in the application footer) to export the current workbook, including every PCI / RSI assignment produced by the solver, ready for deployment and commissioning.

Before exporting

Confirm that all hard validation rules pass, review any remaining soft warnings on the Planning and Clashes tabs, and verify that the Assignments table reflects the intended site / sector configuration.

3.10 Quick reference

Tab	Purpose
Upload & Plan	Load a workbook (Raw = needs planning; Final = already planned) and export the final workbook.
Planning	Top-level PASS/FAIL health view, hard/soft clash counts, interactive network map, per-region breakdown.
Clashes	Colour-coded conflict map, measurement panel, sector explainer, clash inspector, and full clash reference glossary.
Assignments	Searchable / sortable data table of every sector, with CSV export of the current view.
Add Site	Re-plan an existing site or add a brand-new site interactively on the map.
OCTO	Read-only AI agent that answers plain-English questions about the loaded network via LLM reasoning and tool-calling.
Voronoi Compare	Compare an older workbook against the current one and visualise before / after around a chosen site.

Table 4: Quick reference for every section of the 5G Network Planning Portal.

4 Operational Notes

- Use the role assigned to the engineer; role routing determines the default workspace.
- Keep dataset / workbook names descriptive so sessions can be identified later in History.
- Treat AI-generated explanations as engineering assistance; deterministic evidence, KPI values, standards references, and final engineer review remain visible in the workflow.
- Use session history and exported reports to retain an auditable record of analysis and planning actions.